

Technical Appendix

Pipeline

flowchart LR

A["UCSD Goodreads review JSON"] --> B["Sampling by rating and length"]

B --> C["Cleaning and deduplication"]

C --> D["Locked holdout: 1,000 reviews"]

C --> E["No-overlap train/test: 15,000 reviews"]

D --> F["3 independent human labels"]

F --> G["Reliability + consensus"]

D --> H["6-model genAI benchmark"]

H --> I["GPT-4o-mini selected for genAI labeling"]

I --> E

E --> J["Fine-tuned RoBERTa specialist model"]

G --> K["Final holdout evaluation"]

H --> K

J --> K

K --> L["Deploy RoBERTa with GPT-4o-mini escalation"]

Data Files

The final files for modeling are `data/processed/train_no_holdout_overlap.csv`, `data/processed/test_no_holdout_overlap.csv`, and `data/processed/holdout_locked.csv`. The human labels are in `data/labels/holdout_human_labeling_sheet.csv`, and the consensus output is in `data/results/holdout_human_consensus.csv`.

Human Reliability

Dimension	Krippendorff Alpha	Percent Full Agreement
Commitment	0.194	39.4%
Recommendation	0.313	54.6%

Emotion agreement varied by label. The lowest alpha was contentment at 0.045, and the highest was none at 0.380. This supports the project's finding that reader emotions are difficult for humans to align on, especially when reviews express mixed or shifting emotions.

GenAI Benchmark

Supplier	Model	Emotion Macro F1	Commitment Macro F1	Recommendation Macro F1	Cost / 1k	Runtime / 1k	Parse Failures
OpenAI	GPT-4o-mini	0.394	0.516	0.563	\$0.138	2,854 sec	0
OpenAI	GPT-4.1-mini	0.413	0.622	0.711	\$0.409	2,475 sec	0
Anthropic	Claude Sonnet 4.6	0.411	0.618	0.757	\$3.913	3,754 sec	2
DeepSeek	DeepSeek Chat	0.424	0.617	0.720	\$0.269	2,025 sec	0
Grok	Grok 3 Fast	0.451	0.611	0.737	\$0.741	1,971 sec	0
Grok	Grok 4.1 Fast	0.421	0.523	0.704	\$0.164	8,978 sec	0

GPT-4o-mini was selected as the best overall genAI strategy because it had usable performance, the lowest cost, and no parse failures. Other models scored higher on some individual metrics, but GPT-4o-mini offered the best practical balance for scalable labeling.

Specialist Model Experiments

Configuration	Model	Learning Rate	Epochs	Batch Size	Emotion Macro F1	Commitment Macro F1	Recommendation Macro F1
1	roberta-base	2e-5	3	8	0.662	0.804	0.818
2	roberta-base	1e-5	5	16	0.690	0.804	0.816
3	roberta-base	2e-5	5	16	0.713	0.839	0.840
4	roberta-base	3e-5	10	16	0.746	0.826	0.840

Configuration 3 was selected because it offered strong F1 scores without the longer training time and overfitting risk of the 10-epoch configuration.

RoBERTa Holdout Evaluation

Task	Holdout Macro F1	Holdout MCC / Sample F1
Emotions	0.406	Sample F1 0.480
Commitment	0.622	MCC 0.460
Recommendation	0.700	MCC 0.552

RoBERTa outperformed GPT-4o-mini on all three holdout tasks: emotions 0.406 vs 0.394, commitment 0.622 vs 0.516, and recommendation 0.700 vs 0.563.

Error Analysis

The main model weaknesses were rare emotions and overlapping emotional categories. GPT-4o-mini had low F1 for disgust, fear, and anger. Disgust had F1 0.000, fear had F1 0.200, and anger had F1 0.231. The model also over-applied joy, which had very high recall but low precision.

The broader confusion pattern was among joy, love, interest, and contentment. This matches the human reliability issue because those labels are also difficult for people to separate.

Mitigation Plan

Use RoBERTa for routine production classification and keep GPT-4o-mini as a low-cost genAI labeling and escalation tool. Add more examples for rare labels, clarify codebook rules for joy, love, interest, and contentment, tune multi-label emotion thresholds, and route low-confidence rare-emotion cases to human review.